Bartek Ogryczak

Senior Back-end Engineer

pronounced: bar-tech og-ree-chuck

Professional Skills

Seasoned engineer with 25+ years of programming experience, 20 years building and scaling high-traffic distributed services. Self-starter, eager to learn new technologies and programming languages, and take on new challenges

AWS, microservices architecture, Python, Django, Kafka, Redis, Ruby, MySQL, resilience, scalability, K8s and more.

Languages spoken: **English** (fluent), **Spanish** (fluent), **Polish** (native)

Contact info

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Work Experience

Senior Software Engineer II (L5) at *Chime*, *San Francisco, CA*

May 2020 — *present*

Currently on **Application Foundry team**, a back-end team adjacent to infrastructure, building tools enabling our engineers to make their services more resilient and scalable. Workload includes writing application code, but also setting up infrastructure and observability. Significant part of my time is also dedicated to technical writing, such as architectural proposals, documentation, evangelizing best practices, incident rubooks, post-mortem analisis, etc.
Previously I was past of **AWS Migration** team, a cross-functional "tiger" teamp responsible for a succesful migration of the entirety of Chime's infrastructure from RackSpace to AWS EKS and a variety of AWS hosted services.
Recent projects and responsibilities:

- Migrated a variety of different task scheduling systems to K8s CronJobs.
  Key results: Single source of truth for task scheduling previously spread accross 5 different systems.
- Actively developing async intra-service communication Hawker, an in-house messaging bus build on AWS SNS, SQS, Lambda, Kinesis and Google Protobufs.
  Key results: enabled teams to interoperate through well defined protocols, deprecated anti-pattern of using Sidekiq for cross-service comms.
- Leveraged intra-service RPC using Twirp.
  Key results: Standarized, well defined RPC increased teams interoperability.
- Driven implementation, best practices, and adoption of concurrency limiters, rate limiters, and circuit breakers.
  Key results: Blast radii of services has been significantly reduced, eliminating most of the cascading failures. As a result incident recovery was quicker and easier.
- Promoting best practices and implementing shared-nothing architecture, particulary for the Redis databases and caches.
  Key results: Moved Chime away from data-fuse anti-pattern, significantly increasing resilience.

Work Experience (continued)

Senior Software Engineer II at *Eventbrite*, *San Francisco, CA*

Oct 2014 — Apr 2020

Key member of **Backend Foundry squad / Infrastructure team**, responsible for spearheading adoption and implementation of architectural solutions enabling Eventbrite to operate at scale, while helping engineers move faster.
Responsibilities:

- Driven deprecation of hard-coded checks in favor of robust, fully dynamic permissions' system. Worked closely with consumer, internal audit, and security teams to enable company-wide adoption. Developed tools and best practices enabling engineers.
  Key results: enabled consumer teams to serve complex, high value customers.
- Requirements gathering, design and implementation of Cerberus, Eventbrite's new generation Authentication, Authorization and Identification service driving the new permissions system. Coordinated with SRE and BDRE on SLA/SLO budgets & monitoring for this mission critical system.
  Key results: receiving most traffic, Cerberus is the most stable and reliable of Eventbrite's services.
- Design, implementation and enabling adoption of PySOA, in-house SOA bus, which enabled migration of monolith into service oriented architecture.
  Key results: remote service call latency reduced from 20-30ms to 1-2ms, while also enabling multiplexing. About 70% of the monolith code migrated into microservices.
- Developed Dilithium, in-house event-sourcing based on parsing MySQL binlog, using rules to generate high-level events from row level changes. Productionized MVP, making it resilient and scalable.
  Key results: enabled teams to use it with ease, making service robust enough to withstand on-sales generating in excess of 5,000 updates a second, absorbing the peaks into Kafka, thus not overwhelming consumers.
- In Architecture Guild, authored architectural proposals and engineering best practices, driving their approval and adoption throughout the organization

Back-end Developer at *Layar*, *Amsterdam, The Netherlands*

Feb 2011 — Sep 2014

Responsible for Layar's AWS back-ends for mobile Augmented Reality app and web based Layar Creator:

- Designed and implemented scalable architecture for visual search. Used an asynchronous server (Tornado) to scale-out otherwise single-process Visual Search Engine.
- Decoupled services using SQS and Twisted.

Core Developer at *Hyves*, *Amsterdam, The Netherlands*

Jul 2009 — Feb 2011

Back-end engineer on Media Storage Team at the Netherland's then largest social network (10M MAU). In charge of 2 billion image/video files with 300K daily new uploads using in-house solutions on top of commodity hardware. Designed and implemented an algorithm which guaranteed files would be replicated on machines not in physical proximity.
Key results: the new algorithm not only guaranteed more resilience, but also improved load distribution, significantly reducing latency while serving hot files.

Work Experience (continued)

Software Analyst at *Red Trabajar*, *Madrid, Spain*

Feb 2008 — Jul 2009

Complete rewrite of Spain's then biggest job portal. Rebuilt from the ground up using Zend Framework and Oracle DB, while seamlessly migrating from legacy website. Designed and implemented a high throughput mailing system capable of generating and sending 15M individually customized emails nightly.

Software Analyst at *Santander Bank*, *Madrid, Spain*

Jul 2006 — Feb 2008

Integrating and processing data from multiple real-time sources for real time analysis and risk management at Santander's investment banking division.
Key results: Developed a better job distribution/aggregation algorithm used on blade servers, which significantly reduced latency, while at same time reduced total run time 12-fold.

Web developer / Sysadmin at *Melog.com*, *Warsaw, Poland*

1997 — 2005

Linux Sysadmin / generalist. As the company grew, specialized in back-end, working mainly on Oferty.net, a meta-search aggregating data from dozens of real-estate agencies.

Education

MS in Computer Science at *University of Warsaw*, *Warsaw, Poland*

1995 — 2001

Master's specialization in **Distributed Systems.**
Thesis project: USOSweb, student and professor facing component of USOS (comprehensive university course management system). After graduation aided further development as an external consultant. USOS is highly successful, it is actively used by over 50 universities, and has won multiple awards.